

Total No. of Questions: 10]

SEAT No. :

P2972

[6004]-955

[Total No. of Pages : 2

B.E. (Mechanical)

**PRODUCT DESIGN AND DEVELOPMENT
(2015 Pattern) (Semester-II) (402050-C) (Elective-IV)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Neat diagram must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of Calculators is allowed*
- 4) *Assume Suitable data if necessary.*

Q1) a) Explain how you will carry out analysis of gathered information [5]

b) Explain Pugh's concept [5]

OR

Q2) a) Differentiate between product development and product design [5]

b) Explain customer need with suitable example [5]

Q3) a) Explain, morphological analysis with suitable example [5]

b) Explain market segmentation with suitable example [5]

OR

Q4) a) Explain the concept of brainstorming with suitable example [5]

b) How standardization, simplification and specialization are related to product design? Explain [5]

Q5) a) What is force flow diagram? Apply and explain for paper punching machine. [8]

b) Explain following tools in benchmarking [8]

i) Intended Assembly Cost Analysis

ii) Trend Analysis

OR

P.T.O.

- Q6)** a) Define benchmarking. What are objective of benchmarking? [8]
b) Explain function form diagram with suitable example [8]
- Q7)** a) Discuss guideline desing for casting and forging in detail [8]
b) Discuss in detail manufacturing cost analysis [8]
- OR
- Q8)** a) Discuss various regional and global issue of environmental pollution [8]
b) Discuss guideline for desing for environment and materials [8]
- Q9)** a) Explain different elements of product life cycle management in detail [10]
b) Explain the elements of PLM in detail [8]
- OR
- Q10)** a) Explain the role of customers and vendors in implementation of PLM [10]
b) Explain the components of PDM in detail [8]